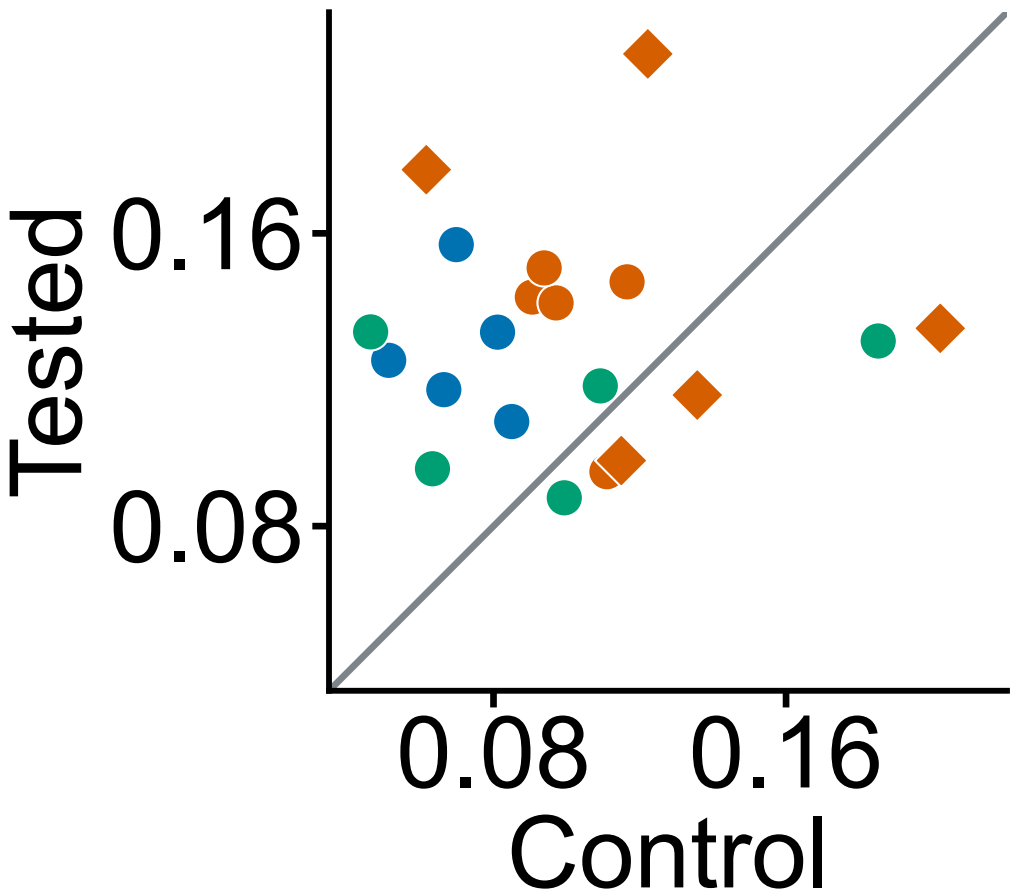DG spatial coding